SCENARIO: “Internet is not working”

If config:

```
vboxuser@Ubuntu:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe7f:18e9  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:a00:27ff:fe7f:18e9  prefixlen 64  scopeid 0x0<glo
bal>
        ether 08:00:27:7f:18:e9  txqueuelen 1000  (Ethernet)
        RX packets 16873  bytes 21673950 (21.6 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 11145  bytes 808069 (808.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 586  bytes 50300 (50.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 586  bytes 50300 (50.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

vboxuser@Ubuntu:~$
```

WHY WE USE IT

To check:

- IP address
- network adapter status
- whether system connected to network

Issue
Internet not working.

Command Used
ifconfig

Findings
Network interface was active and assigned IP address 10.0.2.15.

Conclusion
System successfully connected to network.

SCENARIO:

“Website is not opening”

PING :

```
vboxuser@Ubuntu:~$ ping google.com
PING google.com (142.251.222.110) 56(84) bytes of data.
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=1 ttl=64 ti
me=255 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=2 ttl=64 ti
me=281 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=3 ttl=64 ti
me=275 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=4 ttl=64 ti
me=271 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=5 ttl=64 ti
me=266 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=6 ttl=64 ti
me=273 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=7 ttl=64 ti
me=111 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=8 ttl=64 ti
me=263 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=9 ttl=64 ti
me=262 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=10 ttl=64 t
ime=268 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=11 ttl=64 t
ime=262 ms
```

WHY WE USE IT

Checks:
internet connectivity
DNS resolution
packet communication

Issue
Unable to access websites.

Command Used
ping google.com

Findings
DNS resolution failure detected.

Resolution
Checked DNS configuration.

Learning Outcome
Understood internet connectivity testing.

SCENARIO 3 — CHECK WHETHER INTERNET OR DNS ISSUE:

Ping 8.8.8.8:

```
vboxuser@Ubuntu:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=50.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=48.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=45.7 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=49.9 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=64 time=45.0 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=64 time=36.7 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=64 time=259 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=64 time=47.5 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=64 time=52.8 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=64 time=52.2 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=64 time=45.4 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=64 time=158 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=64 time=55.4 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=64 time=51.3 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=64 time=44.2 ms
64 bytes from 8.8.8.8: icmp_seq=16 ttl=64 time=31.1 ms
64 bytes from 8.8.8.8: icmp_seq=17 ttl=64 time=59.6 ms
64 bytes from 8.8.8.8: icmp_seq=18 ttl=64 time=71.6 ms
```

Diagnostic Test

ping 8.8.8.8

Findings

IP connectivity successful.

Conclusion

Internet connectivity available but DNS resolution failed.

SCENARIO 4 — USER SAYS:

“Need to know system IP”

Hostname -I:

```
vboxuser@Ubuntu:~$ hostname -I
10.0.2.15 fd17:625c:f037:2:a00:27ff:fe7f:18e9
```

SCENARIO 5 — CHECK DNS INFORMATION

```
vboxuser@Ubuntu:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.221.238
Name:   google.com
Address: 2404:6800:4009:810::200e

vboxuser@Ubuntu:~$
```

WHY WE USE IT

Checks:

- DNS server
- domain resolution

OUTPUT MEANING

Address: 142.250.x.x

Means:
DNS converted:

google.com → IP address

Issue

Website hostname not resolving.

Command Used

nslookup google.com

Findings

DNS successfully resolved hostname to IP address.

Learning Outcome

Understood DNS resolution process.

SCENARIO 7 — CHECK OPEN CONNECTIONS:

```
vboxuser@Ubuntu:~$ netstat -tulnp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
-
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
-
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
-
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
-
tcp6       0      0 :::22                   :::*                    LISTEN
-
tcp6       0      0 ::1:631                 :::*                    LISTEN
-
udp        0      0 127.0.0.54:53           0.0.0.0:*
-
udp        0      0 127.0.0.53:53           0.0.0.0:*
-
udp        0      0 127.0.0.1:323           0.0.0.0:*
-
```

WHY WE USE IT

Shows:

- listening ports
- active services
- network connections

SERVICE DESK USE

Useful when:

- application not responding
- server service issue
- port troubleshooting

SCENARIO 8 — CHECK NETWORK PATH:

```
vboxuser@Ubuntu:~$ traceroute google.com
traceroute to google.com (192.178.174.101), 64 hops max
  1   10.0.2.2  0.005ms  0.002ms  0.002ms
  2   *  *  *
  3   *  *  *
  4   *  *  *
  5   *  *  *
  6   *  *  *
  7   *  *  *
```

WHY WE USE IT

Shows:

route packets travel through network.

SERVICE DESK USE

Used when:

- internet slow
- routing problem
- packet loss issue

SCENARIO 9 — REMOTE LOGIN

ssh user@192.168.1.10

WHY WE USE IT

Allows:

remote system access.

SERVICE DESK USE

Very common in:

- Linux support
- server administration
- remote troubleshooting

SCENARIO 10 — COPY FILES REMOTELY

scp file.txt user@192.168.1.10:/home/user

WHY WE USE IT

Securely transfers files between systems.

SERVICE DESK USE

Used for:

- log transfer
- backup
- troubleshooting files